Chapter 1: Classification of Living Organisms

Exercise — Chapter 1: Classification of Living Organisms

i. Which one of the following is the foot of an aquatic bird?

The webbed foot — the foot with skin between the toes, which helps the bird swim. (The options in the book are pictures of feet.)

ii. The cat belongs to which group?

(a) Amphibians

(b) Reptiles

(c) Birds

✓ (d) Mammals

iii. A fish respires through which organ?

✓ (a) Gills

(b) Lungs

(c) Air sacs

(d) Skin

iv. What is the number of cotyledons in a gram seed?

(a) One

✓ (b) Two

(c) Three

(d) Four

v. Which group is found only in the ocean?

(a) Fish

(b) Insects

✓ (c) Echinoderms

(d) Amphibians

2. i. Define biodiversity.

The variety of plants and animals found in a particular place is called biodiversity.

ii. State the importance of classification.

Due to the structure and other characteristics of organisms, we can identify and study them and know the relationships among them. Classification also helps us conserve biodiversity.

iii. (a) Write the differences between worms and insects.

Worms have soft bodies without legs, for example the earthworm. Insects have six legs, three body parts (head, thorax and abdomen) and antennae, for example the ant and the butterfly.

(b) Write the differences between amphibians and reptiles.

Amphibians live both in water and on land, have moist skin and lay their eggs in water, for example the frog. Reptiles live on land, have dry and scaly skin and lay their eggs on land, for example the lizard and the snake.

iv. What is the similarity between fungi and animals?

Both fungi and animals cannot make their own food. They depend on other organisms or dead matter for their food.

v. Write one characteristic and one example of Monera and Protista.

Monera: These organisms are made of one cell and their structure is very simple, for example bacteria.

Protista: Most of them live in water, for example amoeba and paramecium.

3. i. Some animals have an internal skeleton of cartilage such as shark and sting ray. What is the importance of cartilage in their life?

Cartilage is light and flexible. A skeleton of cartilage makes the body of the shark and the sting ray light, so they can swim easily and move flexibly in water.

ii. Why does the body of a zebra have black and white stripes?

The black and white stripes break up the zebra's outline in tall grass and confuse predators, so the stripes help the zebra hide from its enemies.

iii. Why are the ears of an elephant large?

Elephants live in a hot climate. Their large ears help them release body heat and keep cool.

4. Investigate: Investigate the factors responsible for the extinction of biodiversity. Give an example of an extinct and an endangered animal. Investigate what types of animals are most likely to become extinct.

5. Project: Album of Animals — Cut pictures of two animals from each of the five groups of vertebrates, paste them in an album and write the characteristics of each animal with its picture.

Chapter 2: Microorganisms

Exercise — Chapter 2: Microorganisms

i. Mushrooms belong to which group?

(a) Viruses

✓ (b) Fungi

(c) Bacteria

(d) Protozoa

ii. What causes polio?

(a) Protozoa

✓ (b) Virus

(c) Fungi

(d) Bacteria

iii. Penicillium is an example of which group?

(a) Protozoa

✓ (b) Fungi

(c) Bacteria

(d) Viruses

iv. Food is contaminated due to the presence of _____ in the environment.

(a) Moisture

✓ (b) Microorganisms

(c) Air

(d) Heat

v. Which one of the following is not a microorganism?

(a) Bacteria

(b) Virus

(c) Algae

✓ (d) Ant

2. i. What is a pathogen? How does it enter in the bodies of organisms?

A pathogen is a microorganism that causes disease. It enters the bodies of organisms through air, water, food, animals (for example insect bites) and direct contact.

ii. Why are some bacteria and fungi called decomposers?

Some bacteria and fungi break down the complex matter of dead bodies into simple materials. Because they decompose dead matter, they are called decomposers.

iii. Write two benefits and two harmful effects of bacteria.

Benefits of bacteria: They are used to prepare foods like yoghurt, and they decompose dead matter and help make medicines.

Harmful effects of bacteria: They cause diseases like typhoid and cholera, and they spoil food.

iv. How does the microorganism yeast work to soften and rise dough of flour?

Yeast feeds on the sugar in the dough and releases carbon dioxide gas. The gas forms bubbles in the dough, which soften it and make it rise.

v. How does a mosquito transmit disease?

A mosquito bites an infected person and takes the pathogen into its body. When it bites another person, it injects the pathogen into that person's body. For example, the female Anopheles mosquito spreads malaria in this way.

3. The soil of forests of hot climate is always hot and moist, whereas the soil of forests of cold climate is cold and dry. Out of these two areas, where will the decomposition of fallen leaves happen more quickly? Explain.

Decomposition of fallen leaves will happen more quickly in the forest of hot climate. The soil there is always hot and moist, and microorganisms grow and work faster in hot and moist conditions than in cold and dry soil.

4. Investigate: “All microorganisms are harmful and cause diseases”. Prove that this idea is incorrect — for example, bacteria are used to make yoghurt and bread, fungi like yeast are used in baking, and many microorganisms decompose dead matter and are used to make medicines.

5. Many people use antibacterial soap to kill the bacteria present on their hands. But due to excessive use of soap, the chance of getting infection increases instead of decreasing. Why does it happen?

Excessive use of antibacterial soap kills the useful bacteria on the skin as well, and the harmful bacteria that survive can become resistant. This disturbs the skin's natural protection, so the chance of getting an infection increases instead of decreasing.

Chapter 3: Flowers and Seeds

Exercise — Chapter 3: Flowers and Seeds

i. The given figure is of a dicot seed. Which statement is correct about it?

(a) Endosperm is present which stores food.

(b) There is no role of cotyledon in storing food.

✓ (c) Cotyledon stores food for the embryo.

(d) Cotyledon appears in the form of protective cap.

ii. In the given picture of a flower, which are the reproductive parts of the flower?

Options in the book: (a) A, B and D (b) B, C and D (c) A, B and C (d) A, C and D.

The reproductive parts in the picture are A (the anther, the male part) and D (the ovary, the female part). B is the sepal and C is the petal, which are not reproductive parts. Note: the book's printed options are groups of three and none of them equals A and D exactly.

iii. The gram seed is covered by which structure?

(a) Cotyledon

✓ (b) Seed coat

(c) Plumule

(d) Endosperm

iv. Which conditions are necessary for seed germination?

(a) Water, soil and air, darkness

(b) Air, water, light

✓ (c) Water, temperature, air

(d) Temperature, soil, light

v. Which type of pollination is a must for papaya?

(a) Self-pollination

✓ (b) Cross pollination

(c) Both types of pollination

(d) None of these

2. i. Can you tell the names and functions of the four parts of a flower?

The four parts of a flower are: (1) Sepals — green leaflets that protect the internal structures. (2) Petals — coloured leaflets that attract insects and birds. (3) Stamens — the male reproductive part, consisting of the anther (where pollen grains are formed) and the filament (which supports the anther). (4) Carpel — the female reproductive part, consisting of the stigma (where pollen grains adhere), the style and the ovary (which has ovules).

ii. How will you compare self-pollination and cross-pollination?

In self-pollination, pollen grains are transferred from the anther of a flower to the stigma of the same flower or of another flower of the same plant. In cross-pollination, pollen grains are transferred from the anther of a flower to the stigma of a flower of a different plant of the same kind.

iii. Compare gram and maize seed.

The gram seed is a dicot seed: it has two cotyledons, which store food for the embryo. The maize seed is a monocot seed: it has one cotyledon, and it consists of a seed coat and an embryo.

iv. Define reproduction.

Reproduction is the process by which organisms produce new organisms of their own kind for the continuation of their generation.

v. Label the given picture of a flower.

Label the parts of the flower as follows: sepal, petal, stamen (anther and filament), carpel (stigma, style and ovary), ovule and pedicel (the stalk of the flower).

3. i. Observe the given pictures (plain water, water with salts, water with sugar, water with vinegar) and tell what is the role of water in seed germination?

Water is necessary for seed germination. It softens the seed coat and activates the embryo, so the seed can germinate. In the pictures, seeds germinate in plain water, but they do not germinate properly in water with salts, sugar or vinegar, because these disturb germination.

3. ii. Observe the given pictures of flowers. For which type of pollination are the shapes and structures of the flowers suitable? (Answer from the pictures in the book: flowers with large, coloured petals and exposed reproductive parts are suited to cross-pollination by insects.)

4. Investigate: If all the insects become extinct, what will be its effect on flowering plants? (Insects carry out pollination; without them, cross-pollination would stop and many flowering plants could not form seeds.)

5. Project: Collect five different types of flowers. See in which flowers sepals, petals, stamens and carpels are present and in which these are not present. Compare these structures in different flowers. Also make a model of a flower using low-cost or no-cost material.

Chapter 4: Environmental Pollution

Exercise — Chapter 4: Environmental Pollution

i. Which disease is caused due to air pollution?

(a) Diarrhea

(b) Typhoid

✓ (c) Lungs cancer

(d) Cholera

ii. The germs present in _____ cause typhoid.

✓ (a) Sewerage water

(b) Fertilizers

(c) Factory waste

(d) Insecticides

iii. Which one of these is non-biodegradable?

(a) Feathers

(b) Papers

(c) Leaves of plants

✓ (d) Polythene bag

iv. Which of the following is NOT a greenhouse gas?

✓ (a) Oxygen

(b) Methane

(c) Ozone

(d) Carbon dioxide

v. Which one of the following acts as a source of air pollution?

(a) Collecting rubber

✓ (b) Burning rubber

(c) Reusing rubber

(d) Recycling rubber

2. i. Write three causes of air pollution.

Three causes of air pollution: smoke from factories, smoke from vehicles, and burning of rubber, plastic and other wastes.

ii. What are four effects of pollution on life?

Four effects of pollution on life: it causes diseases like lung cancer, typhoid and cholera; it harms plants and animals; it disturbs the balance of nature; and it damages the environment in which we live.

iii. What is greenhouse effect?

The greenhouse effect is the warming of the Earth because greenhouse gases (like carbon dioxide, methane and ozone) trap the Sun's heat in the atmosphere and do not let it escape back into space.

iv. What will be the effects of global warming?

Global warming raises the temperature of the Earth, melts glaciers and ice, raises the sea level, disturbs rainfall and seasons, and harms plants, animals and human life.

v. Write three ways of preventive measures to reduce pollution.

Three preventive measures to reduce pollution: do not burn rubber and plastic; dispose of wastes properly instead of throwing them into water; and plant more trees and use polythene bags less.

3. i. What is the main cause of pollution?

The main cause of pollution is human activity — smoke from factories and vehicles, untreated sewerage and factory waste thrown into water, and non-biodegradable wastes like polythene bags.

ii. Why has the Government of Pakistan banned the use of polythene bags?

The Government of Pakistan has banned the use of polythene bags because they are non-biodegradable: they do not decompose, they block drains, pollute soil and water, and harm animals that eat them.

iii. If it is written D2W on the polythene bags, then why is there no ban on the use of such bags?

Bags marked D2W contain an additive that makes the plastic break down into small pieces after some time, so they are degradable and are not covered by the ban on ordinary non-degradable polythene bags.

4. Investigate: What is the relationship between diseases and pollution? (Polluted air causes lung diseases, polluted water causes typhoid, cholera and diarrhea, and polluted soil and surroundings spread germs.)

Chapter 5: Physical and Chemical Changes of Matter

Exercise — Chapter 5: Physical and Chemical Changes of Matter

i. The change of milk into yogurt is:

(a) Physical change

(b) Climate change

✓ (c) Chemical change

(d) Change of colour

ii. Why did a person paint his iron gate? 1. To save from rusting 2. To save from sunlight 3. To make it beautiful 4. To save from water. Out of these, which answer is correct?

(a) 1 and 2

(b) 1 and 3

(c) 2 and 3

✓ (d) 1 and 4

iii. Which factor will not affect the dissolving of sugar in water?

✓ (a) Adding salt in water.

(b) Making sugar powder by grinding.

(c) Heating water and sugar

(d) Stirring water and sugar

iv. Which one is not a chemical change?

(a) Seed germination

✓ (b) Making paper boat

(c) Burning of wood

(d) Cooking food

v. What type of change is it when metal expands on heating?

(a) Permanent

(b) Chemical

✓ (c) Physical

(d) Temporary

2. i. Explain evaporation giving examples from everyday life.

Evaporation is the change of a liquid into gas (vapours) from its surface, without boiling. Examples from everyday life: wet clothes dry in the Sun, water in a cup decreases when left open, and a wet floor dries after some time.

ii. Define condensation.

Condensation is the change of a gas into a liquid. For example, water vapours in the air condense into drops of water on a cold surface.

iii. What is rusting and which type of change is it?

Rusting is the change that occurs on the surface of iron due to the action of oxygen and water, which forms a brown layer on iron. It is a chemical change, because a new material with new properties is formed.

iv. Give an example of a chemical change in which carbon dioxide is produced.

When vinegar is mixed with baking soda, bubbles of carbon dioxide gas are formed. Burning of wood also produces carbon dioxide.

v. Explain the three states of matter and their interconversion.

The three states of matter are solid, liquid and gas. They can change into one another: a solid melts into a liquid on heating, a liquid evaporates into a gas, a gas condenses into a liquid on cooling, and a liquid freezes into a solid on cooling.

3. i. Weigh one-fourth glass of vinegar and one spoon of baking soda. Mix them together; bubbles will be formed. Weigh this compound again; its weight becomes less than the previous one. How will you explain the loss of weight?
When vinegar and baking soda are mixed, bubbles of carbon dioxide gas are formed and the gas escapes into the air. Because the gas leaves the mixture, the weight of the mixture becomes less than before.

ii. On mixing vinegar and boiling water, bubbles are formed. Out of these, one is a chemical change and the other one is a physical change. Explain.
On mixing vinegar and boiling water, bubbles are formed in both cases, but they are different changes. The bubbles in vinegar are a chemical change, because vinegar reacts and a new gas (carbon dioxide) is formed. The bubbles in boiling water are a physical change, because only water changes into steam and no new material is formed.

iii. A student weighs a piece of ice and then allows it to melt. In your opinion, what will be the weight of the water and why?
The weight of the water will be the same as the weight of the ice. Melting is a physical change: only the state changes from solid to liquid, and no matter is added or lost, so the mass remains the same.

Chapter 6: Light and Sound

Exercise — Chapter 6: Light and Sound

i. How does light travel in air?

(a) In a circle

(b) Along a curved path

✓ (c) Along a straight line

(d) In dispersed form

ii. Which object reflects maximum light?

(a) White paper

(b) Coloured paper

✓ (c) Mirror

(d) Black paper

iii. Speed of sound is maximum in:

✓ (a) A metal wire

(b) Air

(c) Water

(d) Vacuum

iv. Which one of the following is called noise?

(a) Sound of a flute

(b) Rustling of leaves

✓ (c) Tractor

(d) Chirping of birds

v. When water comes in the way of sound travelling through air:

(a) Sound will stop

(b) Sound will slow down

✓ (c) Sound will become fast

(d) No effect on the speed of sound

2. i. What is a transparent object? Write the names of three transparent objects.

A transparent object is one through which light passes completely, so we can see clearly through it. Three examples: window pane (glass), clean water and clear plastic sheet.

ii. When is your shadow the shortest and the longest in sunlight?

Your shadow is the shortest at noon, when the Sun is overhead, and the longest in the morning and evening, when the Sun is low in the sky.

iii. The Moon is non-luminous like our Earth. How does it look luminous to us?

The Moon is non-luminous: it has no light of its own. It looks luminous to us because it reflects the light of the Sun towards the Earth.

iv. Can you be a winner while running with your shadow? Give the reason also.

No, you cannot be a winner while running with your shadow, because the shadow moves with you. As long as you and the source of light move together, the shadow stays with you and you cannot leave it behind.

v. When a train is moving away from you, will the intensity of its sound increase or decrease?

When a train is moving away from you, the intensity of its sound will decrease, because the sound has to travel a longer distance to reach you and it spreads out.

Chapter 7: Electricity and Magnetism

Exercise — Chapter 7: Electricity and Magnetism

i. The flow of current in an electric circuit is controlled by:

(a) Bulb

(b) Copper wire

✓ (c) Switch

(d) Rubber

ii. Current can easily pass through:

✓ (a) Steel paperclip

(b) Glass strip

(c) Plastic comb

(d) Wooden spoon

iii. How many poles are there on a ring-shaped magnet?

(a) 1

✓ (b) 2

(c) 3

(d) 4

iv. A freely suspended bar magnet always settles in:

(a) East-west direction

✓ (b) North-south direction

(c) Any direction

(d) Oscillating

v. Which one is a true statement?

(a) North pole attracts north pole

✓ (b) North pole repels north pole

(c) South pole repels north pole

(d) South pole attracts south pole

2. i. Why does a bulb light up in a circuit?

A bulb lights up in a circuit when the circuit is complete: current flows from the cell through the wires and the filament of the bulb, the filament becomes hot and glows.

ii. You are given a bulb. What other components are required to light it up?

To light a bulb you also need a cell (battery), connecting wires and a switch, so that a complete circuit can be made.

iii. Why does an iron nail stuck to a magnet attract another nail?

An iron nail stuck to a magnet becomes a magnet itself for some time (magnetism is induced in it), so it attracts another nail.

iv. How are directions known using a magnetic compass?

A magnetic compass has a freely suspended needle which always points in the north-south direction. The end pointing north shows the north direction and the other end shows the south direction; the other directions are found from these.

v. A balloon is charged by rubbing it with a woollen cloth. Why does it stick to a wall when brought near it?

When a balloon is rubbed with a woollen cloth, it gets a static electric charge. When this charged balloon is brought near a wall, it attracts the opposite charge on the wall's surface, so it sticks to the wall.

3. Two ring magnets are shown on a wooden stand. The upper magnet rises up and is suspended in air, as shown in the figure. a. Why does the upper magnet float in the air? b. Label the N and S poles of the magnets.

The upper magnet floats in the air because the like poles of the two magnets face each other, and like poles repel. The force of repulsion balances the weight of the upper magnet, so it stays suspended.

Label the poles: the facing ends of the two magnets are like poles (for example both north poles face each other), and the outer ends are the opposite poles.

Chapter 8: Structure of the Earth

Exercise — Chapter 8: Structure of the Earth

i. The outer layer of the Earth is called:

(a) External core

(b) Mantle

✓ (c) Crust

(d) Inner core

ii. What are the main types of the soil?

(a) Silt, sand and rock

(b) Clay, humus and sand

✓ (c) Sand, silt and clay

(d) Silt, humus and clay

iii. Which type of soil is good for growth of plants?

(a) Sand

✓ (b) Silt

(c) Clay

(d) Rock

iv. What is the role of living organisms in the soil?

✓ (a) Making soil soft and airy

(b) Making soil hard

(c) Making soil unfit for cultivation

(d) Making the air dry

v. The amount of fresh water on the Earth is:

✓ (a) About 1%

(b) About 30%

(c) About 70%

(d) About 97%

vi. What is important in the natural manure?

(a) Minerals

✓ (b) The decayed matter

(c) Sand

(d) Salts

2. i. What is the outer layer of the Earth called? What is its importance in our life?

The outer layer of the Earth is called the crust. It is important in our life because we live on it: our houses, fields, roads and all land life exist on the crust, and the soil on it gives us food.

ii. Which soil is useful to make pottery? What are its properties?

Clay is useful to make pottery. Its properties: it is soft and sticky when wet, so it can be given any shape, and it becomes hard when it dries.

iii. Which matter is obtained from decayed plants and insects?

Humus is the matter obtained from decayed plants and insects. It makes the soil fertile.

iv. What part of the Earth's surface consists of oceans? Why can we not drink sea water?

About 71% of the Earth's surface consists of oceans. We cannot drink sea water because it is salty (about 97% of the Earth's water is salty) and is not fit for drinking.

v. Where is water found in the form of solid, liquid and gas (vapours)?

Water is found in all three states: as a solid in glaciers and ice, as a liquid in oceans, rivers, lakes and springs, and as a gas (vapours) in the air.

3. i. Label the diagram of the water cycle.

Label the water cycle as follows: water evaporates from oceans, rivers and lakes by the Sun's heat; the vapours rise and condense into clouds; the clouds release water as rain or snow (precipitation); and the rain water flows back into water bodies (collection).

ii. Which soil is used to plaster the houses? Which property of the soil is useful to do so?

Clay is used to plaster the houses, because it is sticky and soft when wet and becomes hard and strong when it dries.

iii. How do the plant roots and other creatures inside the Earth breathe?

Plant roots and creatures inside the Earth breathe through the air present in the small pores (spaces) between the soil particles.

iv. Estimate the piece of silt (fertile soil) from the given figure.

The given figure shows the layers of soil: the top layer contains humus and small plants, below it is a layer of sand and other minerals, and the fertile (silt) part is the layer good for cultivation.

4. Investigate: You have heard of the devastation caused by floods. What is its reason? How can the surplus water of floods be made useful? (Floods are caused by heavy rain and overflowing rivers; the surplus water can be stored in dams and reservoirs and used for irrigation and drinking.)

Chapter 9: Space and Satellites

Exercise — Chapter 9: Space and Satellites

i. Countless shining lights seen at night in the sky are:

✓ (a) Stars

(b) Moons

(c) Planets

(d) Different heavenly bodies

ii. How long does it take for a geo-stationary satellite to complete one orbit?

✓ (a) One day

(b) One week

(c) One month

(d) One year

iii. First man to step on the Moon?

(a) Edwin Aldrin

✓ (b) Neil Armstrong

(c) Yuri Gagarin

(d) Alan Shepard

iv. The system that locates the position of an object on the Earth's surface is:

(a) GRS

(b) GPS

(c) PGS

(d) GSP

v. The first artificial satellite was sent into space in:

(a) 1957

(b) 1959

(c) 1969

(d) 1973

2. i. Write five benefits which we get from artificial satellites.

Five benefits of artificial satellites: they help in communication (telephone, television and internet); weather forecasting; navigation through GPS; exploring the Earth and space; and defence and disaster management.

ii. The Moon is the natural satellite of the Earth. Is the Earth also a satellite of some bigger object? What is that object?

Yes, the Earth is also a satellite of a bigger object: the Sun. The Earth revolves around the Sun, just as the Moon revolves around the Earth.

iii. Why does a geo-stationary satellite appear to be stationary?

A geo-stationary satellite appears stationary because it completes one orbit in one day (24 hours), exactly the same time in which the Earth completes one rotation. So it stays over the same place on the Earth all the time.

iv. Why do astronauts wear a space dress? What are its benefits?

Astronauts wear a space dress (space suit) because there is no air, very low pressure and extreme temperature in space. The space suit provides oxygen to breathe, maintains suitable pressure and temperature, and protects them from harmful rays and small space particles.

v. Why can a telescope installed on a satellite take clearer photographs than from the Earth's surface?

A telescope installed on a satellite takes clearer photographs because there is no atmosphere (air, clouds and dust) in space to disturb the light. On the Earth's surface, the atmosphere blurs the images.

Chapter 10: Technology in Everyday Life

Exercise — Chapter 10: Technology in Everyday Life

i. We check the horizontal surface level by using a:

(a) Plumb line

(b) Metre rule

✓ (c) Spirit level

(d) Triangle

ii. What does a mason use to keep a wall vertical?

(a) Metre rule

(b) Spirit level

(c) Triangle

✓ (d) Plumb line

iii. Saline water is used for:

✓ (a) Cleaning wounds

(b) Drinking by patients

(c) Washing hands

(d) Taking a bath

iv. Sterilized gauze is used:

(a) For checking blood pressure

✓ (b) As bandage on wound

(c) For finding the body temperature

(d) For cleaning workplace

v. The emergency phone call number is:

(a) 1122

(b) 2211

(c) 1133

(d) 0911

2. i. What do you need to make a model of a footbridge?

To make a model of a footbridge you need: cardboard or chart paper, sticks or reeds, glue, string, scissors and colours, as given in the chapter's activity.

ii. What is the difference between the functions of spirit level and plumb line?

A spirit level is used to check whether a surface is horizontal (level), while a plumb line is used to check whether a wall or line is vertical (straight up and down).

iii. What is an LED?

An LED is a Light Emitting Diode: a small electronic device that gives light when electric current passes through it. It uses very little electricity and lasts long.

iv. You have to install a picture frame on the wall. How can it be kept straight horizontally as well as vertically?

Use a spirit level to keep the frame straight horizontally, and a plumb line to keep it straight vertically. Adjust the frame until the bubble of the spirit level is in the centre and the plumb line hangs straight along the side.

v. What is meant by first aid?

First aid is the immediate help given to an injured or sick person before proper medical treatment arrives, for example cleaning a wound, stopping bleeding and bandaging it.

3. i. What sort of tools and equipment are required to make a wooden model?

To make a wooden model you need tools and equipment like a saw, hammer, nails, glue, measuring scale, pencil and sandpaper, along with safety precautions while using them.

ii. How does a musical instrument produce sound?

A musical instrument produces sound by vibration: when a part of the instrument vibrates (for example a string, a drum skin or air in a pipe), it makes the air around it vibrate, and these vibrations reach our ears as sound.

iii. Why is it necessary to join long wires of LEDs and short wires together while preparing a light string?

Long wires of LEDs and short wires must be joined together while preparing a light string so that all the LEDs are connected in the correct order (positive to negative) and the circuit is complete, otherwise the lights will not glow.

4. Investigate: You are provided a 6 volt battery, one strip of cardboard and 12 blue colour LEDs. How will you make a light for decoration? (Connect the LEDs in a proper series/parallel circuit on the cardboard strip with the battery, joining long wires to short wires, and test the light.)

5. Project: Make a model of a cot (charpai) by using two reeds of different thickness, glue, knife, steel pins and strings.